

SCIENTIFIC ENQUIRY SKILLS

BOOK 2

Cheryl Jakab



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resources



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Cheryl Jakab (M.Sc.B.Ed.) has been involved in science education for over 30 years as a teacher, researcher and writer. After completing her Science Masters in Pathology she worked as a lecturer in teacher education mainly in the areas of Science and Technology. Her interest in writing and editing teacher resource materials developed through involvement in post graduate education of practising teachers. She left full time teaching in 1997 to concentrate on a research, consultancy and writing career. Since her first teacher resource series 'Exploring together' was published in 1988 Cheryl has written many teacher and student texts in Science and Technology including contributions to the Curriculum Corporation's secondary science series 'Science Australia'. She has authored numerous children's reading texts, library series, a science and technology dictionary and a soon to be published science encyclopedia for middle years students. Cheryl is committed to developing science curriculum materials that encourage the curiosity of the learner and acknowledge that each learner has their own interests and abilities that they use to actively construct their knowledge and understandings. Cheryl finds she is constantly learning from children, particularly when working on science talent search and science fair projects.

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INTRODUCTION

Scientific Enquiry Skills is designed to help students develop skills in processes of scientific investigation. Each book contains 43 photocopiable activity sheets suitable for use with 10-14 year olds. The activities complement and support aspects of any existing science program and are divided into the following areas:



Living world (biology)



Matter (chemistry)



Physical world (physics)



Earth (geology) and space (astronomy)

The activities are further organised into the major investigative processes students will practise when using the sheets.

Section 1 Planning investigations

Section 2 Obtaining and presenting evidence

Section 3 Comparing evidence and interpreting results

Section 4 Evaluating investigations

Section 5 Terms, ideas and evidence in science

The grid on pages 4-5 summarises the relevant field of science and content of each sheet along with further details of the scientific investigation processes that form the focus of the sheet.

Activity sheet format

The activity sheets include introductions, instructions and questions along with spaces for students to record answers to questions, create graphs or complete data tables. Most sheets include an extension activity for higher achievers or early finishers.

The activities

Many activities centre on discussions between student characters to create an issue, need or problem to be explored. Each activity is self-contained so that students can use them with minimal support from the teacher. The sheets are suitable for individual students, pairs or small groups of students.

Using the activities

The activities are designed to be used when teachers identify a need to focus attention on particular aspects of scientific enquiry. They should be selected according to individual or group needs.

CONTENTS

Page	Title	Field of science and content area of the topic	Details of skills in scientific enquiry process
Section 1 Planning investigations			
6	Heart rate and fitness	Living world Changes in exercise	Decide on extent and range of data to be collected
7	Breathing out oxygen?	Living world Respiration	Turn ideas into a form that can be investigated
8	Finding your blind spot	Living world Structure and function	Decide on appropriate approach to investigation
9	Earthworm reactions	Living world Animal behaviour	Consider key factors in planning investigations
10	The best bubble mixture	Matter Effects of ingredients	Decide on techniques, equipment and materials for investigation
11	Growing crystals	Matter Crystal formation	Decide on techniques, equipment and materials for investigation
12	Force fields	Physical world Magnetism	Decide on extent and range of data to be collected and make predictions
13	The speed of sound	Physical world Sound	Make predictions about what tests will show
Section 2 Obtaining and presenting evidence			
14	Orders of insects	Living world Classification	Make and justify predictions
15	Classifying critters	Living world Characteristics of living world	Isolate and control variables
16	A close look at beetles	Living world Identifying characteristics	Interpret observations
17	Nightwear flammability	Matter Flammability	Isolate variables and interpret observations
18	States of matter	Matter Particle theory of matter	Use tables and charts to represent data
19	Saturated solution	Matter Solutions	Consider key factors in collecting evidence
20	Refracting light	Physical world Refraction	Use tables and charts to represent data and isolate variables
21	Uses of gears	Physical world Gears	Make appropriate observations and isolate variables
22	Plotting ocean depths	Physical world Reflection of sound	Choose and use appropriate equipment safely
23	Hardness of rocks	Earth and space Comparing rocks	Make appropriate observations to get reliable results
Section 3 Comparing evidence and interpreting results			
24	Big head	Living world Growth	Use tables and charts to represent data appropriately
25	Beaks and feet	Living World Animal structure/function	Use observations and other data to draw conclusions
26	Dinosaurs alive	Living world Fossil evidence	Communicate quantitative data using appropriate diagrams
27	How does an eye focus?	Living world Light and sight	Identify patterns and relationships in data
28	Surface tension	Matter Nature of substances	Use observations, evidence or data to draw conclusions

CONTENTS {CONTINUED}

Page	Title	Field of science and content area of the topic	Details of skills in scientific enquiry process
29	Gear ratios	Physical world Transferring forces	Interpret data, draw conclusions and consider if evidence is sufficient to draw conclusions
30	Testing parachutes	Matter/Physical world Balancing forces	Use scientific knowledge to explain and interpret data
31	Magnetism and electricity	Physical world Electricity	Use scientific knowledge to explain and interpret data and make predictions
32	Flight and force	Physical world Forces	Use scientific knowledge to explain and interpret data
33	Seasons and sunlight	Earth and space Changes during the year	Use scientific knowledge to explain and interpret data
34	How fossils form	Earth and space Geological processes	Consider if evidence is sufficient to draw conclusions
Section 4 Evaluating investigations			
35	Light and growth	Living world Growth of plants	Identify patterns in data and suggest improvements to tests
36	The shape of salt	Matter Crystal formation	Consider and explain anomalies in controlled tests
37	Viscosity of lubricants	Matter Fluids and temperature	Consider if evidence is sufficient to draw conclusions
38	Measuring volume	Physical world Volume	Consider if evidence is sufficient to support conclusions and suggest improvements
39	Electromagnet strength	Physical world Magnetism	Interpret data, draw conclusions and consider if evidence is sufficient to draw conclusions
40	Toy car ramp	Physical world Motion	Use tables and charts to represent data appropriately to consider anomalies in observations and try to explain them
Section 5 Terms, ideas and evidence in science			
41	Microbe image problem	Living world Bacteria	Use tables and charts to represent data appropriately to consider anomalies in observations and try to explain them
42	What is a dinosaur?	Living world Classification	Relationship between questions, evidence and scientific ideas
43	What is entomology?	Living world Classification	Develop shared understanding of scientific terminology
44	Sound and light ideas	Physical world Energy	Role of experimentation, evidence and creative thought in development of scientific ideas
45	Reading electric circuits	Physical world Electric circuits	Develop shared understanding of scientific terminology
46	Conductors and insulators	Physical world Flow of electricity	Relationship between questions, evidence and scientific ideas
47	Continental drift ideas	Earth and space The changing Earth	Develop shared understanding of scientific terminology
48	Earth's tectonic plates	Earth and space The changing Earth	Relationship between questions, evidence and scientific ideas



HEART RATE & FITNESS

A measure of cardiovascular fitness is the rate by which heart rate returns to normal after exercise. A group of Year 8 students wants to investigate how quickly their heart rates return to normal after exercise.

They discuss what measurements to take.

1. When and how many beats should the students count to get their heart rate recovery readings?

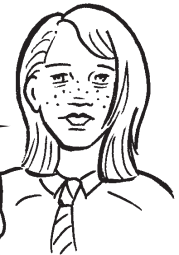
2. What exercise should they do?

3. a. How long should the students measure for?

- b. How will they know it is long enough?

4. How many times should they do the tests to draw any conclusions?

We will need to count all the beats for five minutes after exercise.



No, it would be better to take a measurement for five seconds each minute and multiply by 12.



How long will we exercise for and what will we do?



We should take the heart rate for three minutes after the exercise.



No, I think we will need to measure for longer than that.



What other aspects of fitness could the students investigate as well as cardiovascular fitness? Design a test to investigate one of these.

BREATHING OUT OXYGEN?



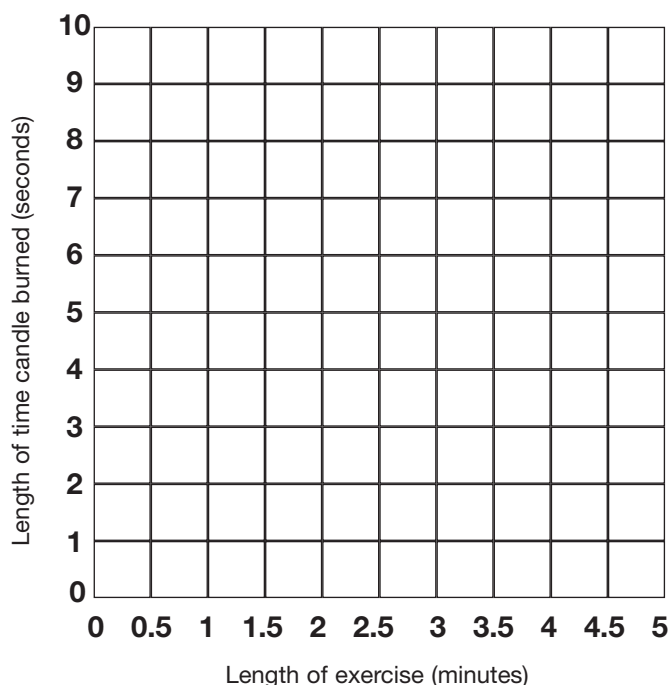
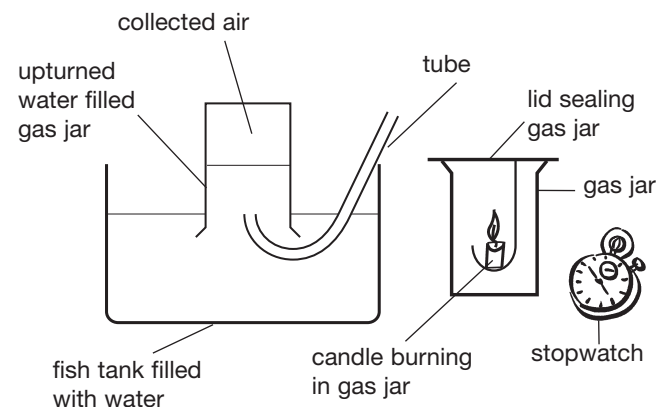
During exercise, an increased depth and rate of breathing supplies extra oxygen to the body. Nabeel and Tanya wonder if any unused oxygen is exhaled with each breath. They plan a way to measure oxygen in exhaled breath at rest and during exercise.

1. How does the apparatus help them show the amount of oxygen breathed out?

2. Are there any ways the experiment design can be improved?

3. They obtain the following table of results after different amounts of exercise. Draw a graph of their results.

Length of exercise (minutes)	Length of time candle burned (seconds)
0	6.5
1	2.9
2	1.5
3	3.5
4	8.6
5	6.5



4. What can they conclude from their graph?

5. Can you tell if any results are anomalous from this data?

6. These results were obtained from testing one person. How reliable are they?

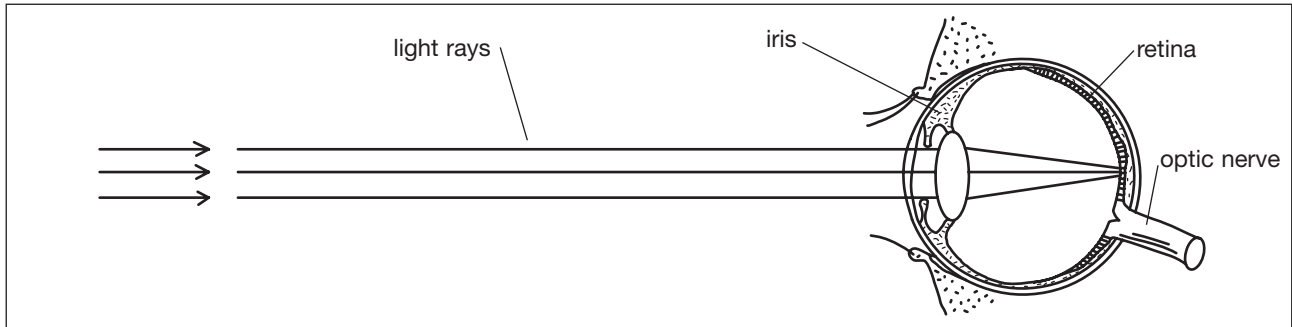


Use a diagram of the respiratory system to help explain how some oxygen is breathed out with each exhalation.



FINDING YOUR BLIND SPOT

Vision receptors in the retina of the eye respond to light that hits them. An area on the retina where the optic nerve is connected to the eye has no vision receptors: this is known as the blind spot.



George and Gyula are exploring a set of instructions which describe how to locate their blind spot.

1. Conduct a preliminary test to see if you can find your blindspot. Describe what it is like when you find it.

X



To find the blind spot:

- Cover the left eye and look at the dot with the right eye.
- Now move your head towards the paper still looking directly at the dot.
- Stop at the point where you cannot see the X.

Gyula and George wonder why they have not noticed their blind spot before. They think that having two eyes is relevant.

2. How does having two eyes reduce the blindspot problem?

3. What would happen if the optic nerve was in the middle of the retina?



Gyula and George wonder whether they will find the blindspot if they change the distance between the dot and the cross, or change their sizes. Design a test to see if this arrangement is the only way to demonstrate the blindspot.

EARTHWORM REACTIONS



Alisa wants to investigate the reactions of worms to touch. She designs two experiments and a results table. Her class then considers how well she planned the experiments to be able to draw conclusions from them.

Experiment 1

Place a dry paper towel on one side of a pan and a moist paper towel on the other. Place a worm across the towels with its head end on the dry towel and observe its response. Repeat with the moist towel.

	Response (moves toward wet or dry)
Head starts on dry	
Trial 1	
Trial 2	
Trial 3	
Trial 4	
Head starts on wet	
Trial 1	
Trial 2	
Trial 3	
Trial 4	

- Why is Alisa using two different starting positions?

- Why does she plan four trials of each situation?

- What conclusions can she draw from her experiment?

- What do you predict will happen?

Experiment 2

Remove the dry paper towel from the pan and let the worm just move around for a while. Use a pipette to place a water droplet on its anterior (front) end and then one on its posterior (back) end. Record its reactions.

Response to water drop on anterior end	
Response to water drop on posterior end	

- What do you predict will happen in these tests?

- What is the difference between the two tests?



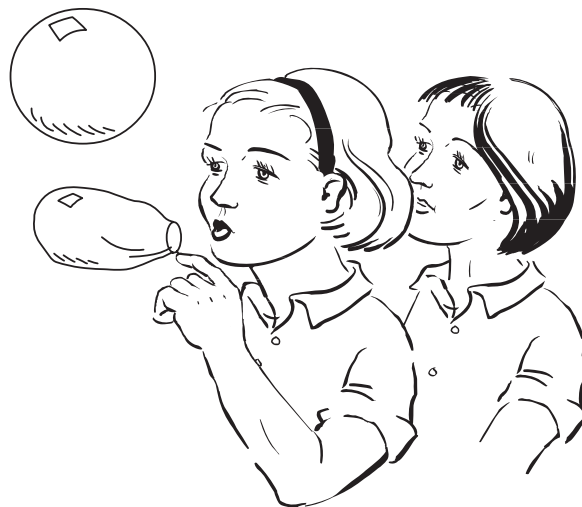
What senses do you predict are important to a worm in its natural habitat?



THE BEST BUBBLE MIXTURE

A soap bubble forms when air is trapped within a film of soap solution. A mixture of approximately 100 parts water to 10 parts detergent to 1 part glycerine is recommended for making bubble mixture.

Emily and Erin plan an investigation to work out the effect of glycerine on making bubbles.



1. What will the glycerine do in the mixture? Make a prediction.

2. What can you do to test this?

3. What are some things that need controlling to make the test fair?

4. What will a 'best bubble mixture' do? Develop your own criteria for a best bubble mixture.

5. Do you think different detergents alter the effectiveness of bubble mixture? How can you test this?

6. Surface tension pulls the bubble into the smallest surface area, which is usually an almost perfect sphere. Suggest how you think you could make different shaped bubbles.



In zero gravity in space, liquids that escape from containers float through the air. Use your knowledge of surface tension to explain what makes them form into balls of liquid.

GROWING CRYSTALS

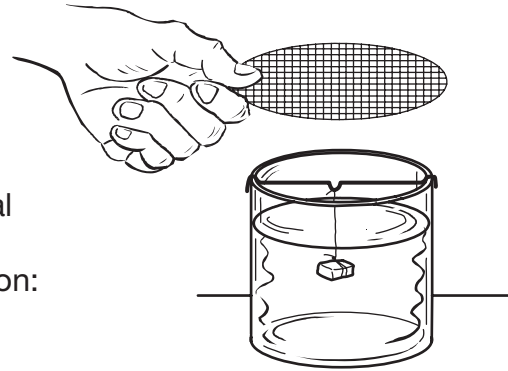


Crystals will grow spontaneously in a saturated solution as the water evaporates. To control the process and grow a large crystal, you can start with a small seed crystal. This leads to the crystal material depositing on the seed rather than haphazardly everywhere in the jar.

Imogen and Phillipa want to grow some crystals for their science fair project. They find a simple set of instructions to follow.

How to grow a crystal

1. Clean a wide jar very well and place a wire over it.
2. Fill the jar with saturated solution.
3. Using thin nylon thread (or fishing line or a hair), tie a string to a seed crystal. You may need to scratch some small grooves into the seed crystal for the string to hold onto the seed.
4. Suspend the seed crystal in the saturated solution: close to, but not on the bottom of the jar.
5. Cover the opening of the jar.



Their teacher says that sometimes a whole carpet of tiny crystals grows on the bottom of the jar, rather than a good large crystal. Help Imogen and Phillipa understand the importance and difficulties of each step by answering the following questions.

1. What happens if the jar is not cleaned well enough?

2. If the thread hangs on the side of the jar instead of on a wire across the top, what is likely to occur?

3. What may happen if they use cotton rather than nylon thread?

4. How important is using a clean, saturated solution?

5. Where is the best place to put the jar after it is set up?

6. What effect will adding more water to the jar have on the growing crystal?



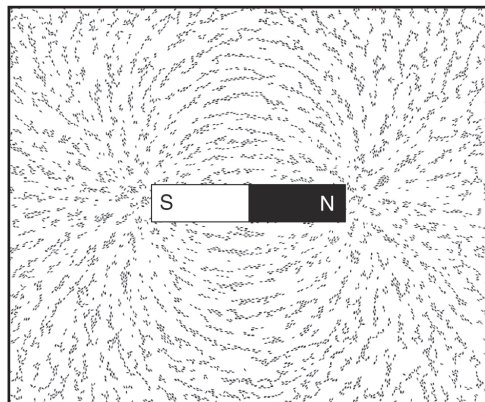
You can try growing your own crystal using this set up. Investigate what chemicals may be best to use.



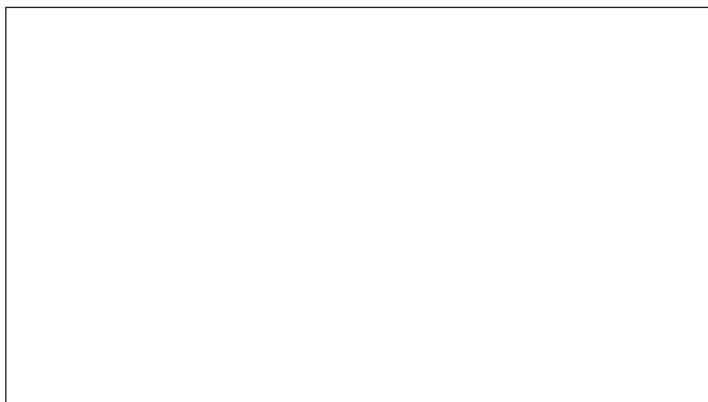
FORCE FIELDS

A magnet does not need to touch an object to move it. Magnetic forces can act at a distance. Magnets are described as having magnetic force fields around them.

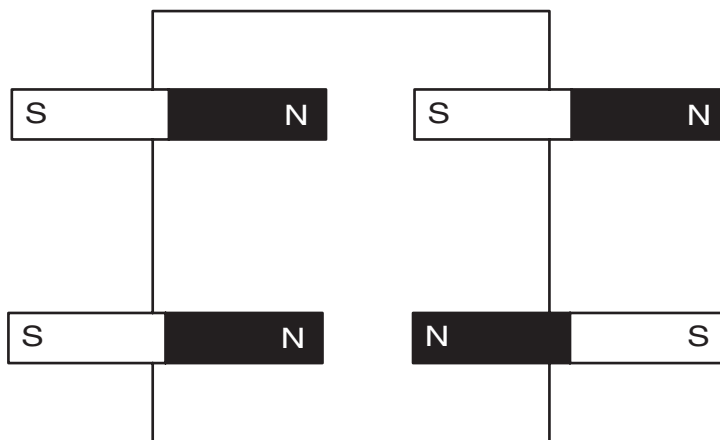
Trish and Emilia are exploring magnetic force fields using iron filings. Their teacher says not to sprinkle the filings directly onto the magnet because they will stick to the magnet in clumps and be very difficult to move.



1. Trish and Emilia need a way to show this force field around their bar magnet. Draw one way they can set up their tests to overcome this problem.



2. Trish and Emilia want to know what happens to the force fields when the magnets are brought together. Explain what they need to do to show this.



3. Predict what will happen to the force fields. Use iron filings or compasses to help you find out if your predictions are correct.



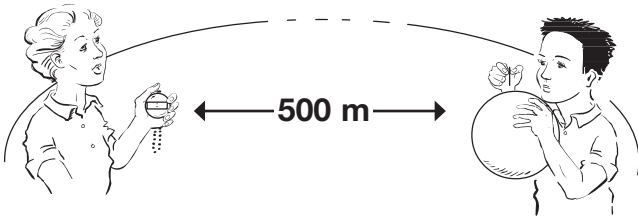
The earth is sometimes described as a giant magnet. Design a way to demonstrate this using a bar magnet.

THE SPEED OF SOUND



In a thunderstorm the flash of lightning is seen before the sound of thunder. Light travels so fast that you see the flash almost as it happens. Sound takes some time to arrive because it travels much slower than light.

A Year 8 class is asked to come up with ways to measure the speed of sound. Ellen and Zoe suggest the following procedure.



Procedure

1. Fill a balloon with flour and blow it up.
2. Measure a distance of 500 metres on the sports oval.
3. With a stop watch, Ellen will stand at the opposite end of the 500 metres to Zoe, who will have the balloon.
4. Zoe will hold up the balloon and prick it with a pin so the flour bursts out in a cloud.
5. Ellen will start the stopwatch when she sees the flour and stop it when she hears the bang of the balloon.

1. What are some things that Zoe and Ellen need to check to ensure their plan will work?

2. What are some things that Zoe and Ellen can do to make this a fair test?

3. What are some problems they may encounter with the test?

4. How many times will they need to do the test?

5. How accurate are their results likely to be?

6. How might they change the test to get a better result?

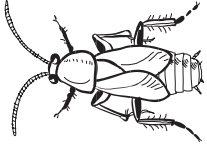
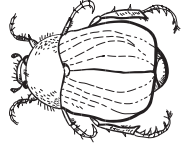
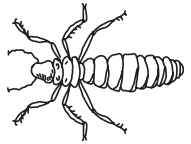
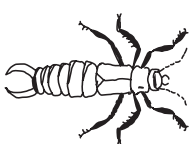


Sound travels in air at about 344 metres per second. About how many kilometres away is a storm, for every second between the lightning and the thunder? If a thunderclap is heard five seconds after a lightning bolt, how far away is the storm?



ORDERS OF INSECTS

Insects are invertebrate animals with six legs and three body parts known as the head, thorax and abdomen. Some insects have wings and some do not. Insects are grouped into orders according to their wing and body structures.

 cockroach – orthoptera	 beetle – coleoptera	 dragonfly – odonata
 termite – isoptera	 earwig – dermaptera	 true bug – hemiptera
 wasp – hymenoptera	 fly – diptera	 grasshopper – orthoptera

A Year 8 class is using a chart to identify the order of some insects they have found.

1. Specimen A has six legs and four wide-veined wings and a long body.
Specimen A may be a _____
because _____
2. Specimen B has six legs and a very narrow waist between the thorax and abdomen.
Specimen B may be a _____
because _____
3. Specimen C has six legs and four very large, patterned wings covered with scales.
Specimen C may be a _____
because _____
4. Specimen D has eight legs and two body parts.
Specimen D is a _____
because _____



Devise a chart with written comments about each group to help distinguish between these insect groups.

CLASSIFYING CRITTERS

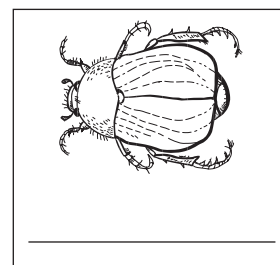
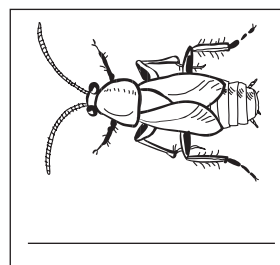
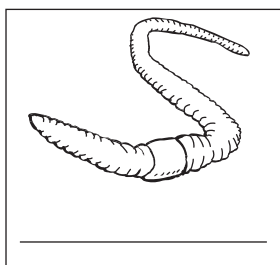
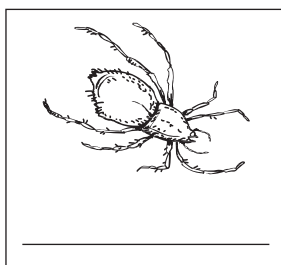
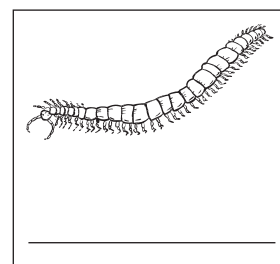
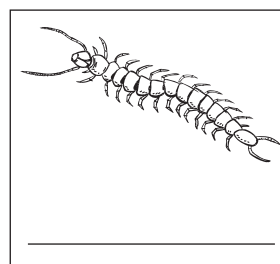
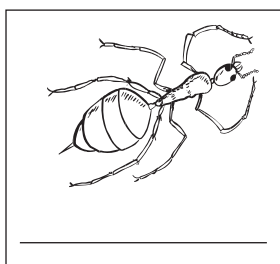
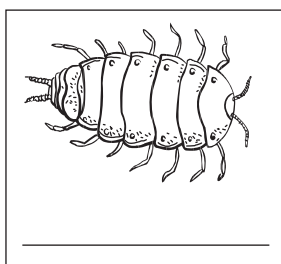


Students in a Year 8 class find a wide range of invertebrate animals in a variety of habitats in the school garden. Izzy and Trish are given the task of classifying the arthropods – animals without a backbone that have legs.

1. Fill in the gaps to complete the chart of arthropod characteristics and give an example of each group.

Class	Number of body sections	Number of pairs of limbs	Number of feelers	Example or common name of class
Crustacea	2	5-7 pairs		
Myriapoda	Head plus many	1 or 2 pairs per segment		
Archnida		4 pairs		
Insecta				

2. To which class do you think these 'crawly critters' belong?



3. Are any of the critters shown not arthropods? How do you know?

4. What is an arthropod? Write what you think the term 'arthropod' means.



Arthropods are invertebrate animals but not all invertebrates are arthropods. Name some invertebrate animal groups that are not arthropods.

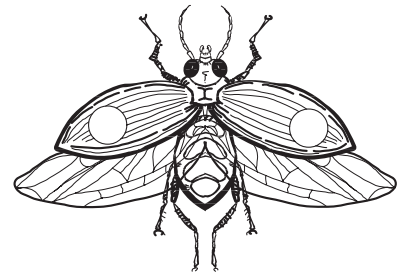


A CLOSE LOOK AT BEETLES

A distinguishing feature of beetles is their elytra – a hard exoskeletal covering over their wings. All beetles have four wings but only use one pair to fly. The other pair of ‘wings’ is a hard covering called an elytron. This is held up when the beetles fly.

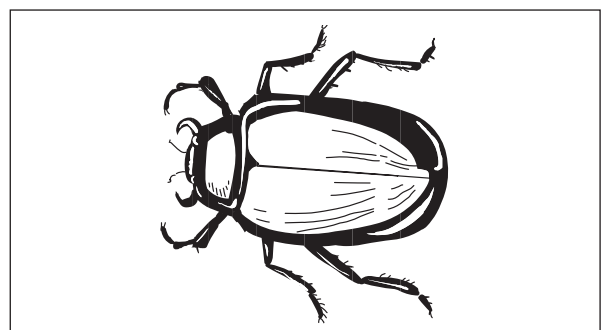
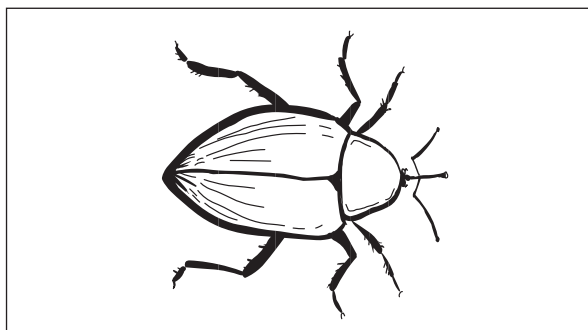
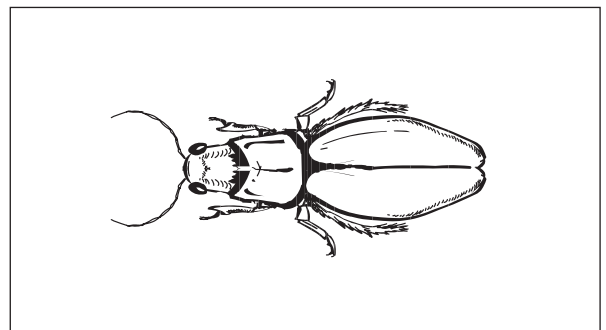
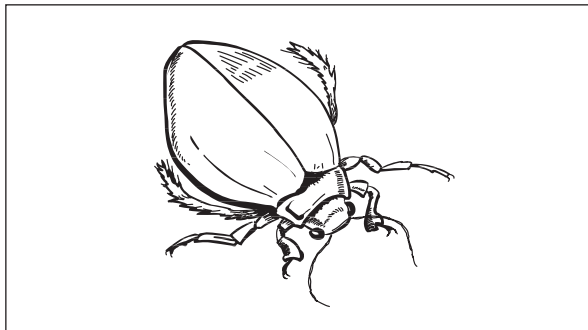
Han and Charlotte collect two specimens from a garden that may be beetles.

1. How do you know if an insect is a beetle?



2. Suggest some ways Han and Charlotte can find out if their specimens are beetles.

3. Observe the four beetles below and then describe their main features in your work book. Use the differences between the beetles to design a key to tell them apart.



It has been suggested that one out of every four or five living animals on Earth is a beetle. Beetles belong to the order Coleoptera, the largest order of living things by number of species. Find out how many beetle species have been identified by scientists.

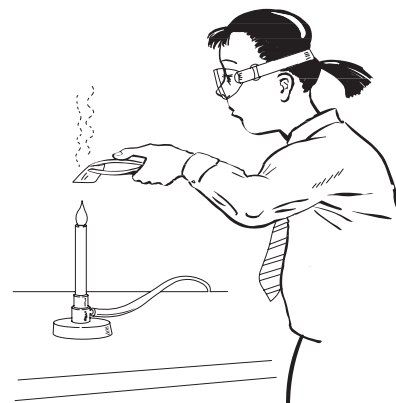
NIGHTWEAR FLAMMABILITY



Clothing catching fire is a major cause of burns in children. Labels on nightwear must contain information about a garment's fire safety level. Materials can be rated for fire safety or danger according to the way they burn.

Janet and Kimberley use the apparatus in the diagram (right) to rank some different fabrics for fire safety. They record how long it takes for six different fabrics to burn away completely.

Material	Time (seconds)
Cotton	12
Silk	5
Wool	14
Nylon	3
Polyester	4
Poly/cotton	7



1. Kimberley explains to Janet that they need same-size pieces of material for a fair test. How do you think size would affect the fairness of the test?

2. What are some other factors they need to control so they can draw conclusions from their tests?

3. List three other observations or tests they can do to compare the fire safety of the fabrics.

4. What might Janet and Kimberley deduce from their test?

5. What safety precautions do they need to consider in carrying out their tests?



Describe how the design of nightwear can affect fire safety. What makes fire safety such an important issue for nightwear in particular?

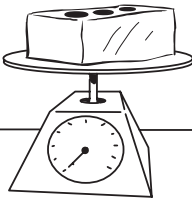
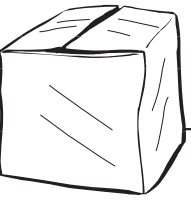
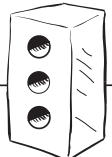


STATES OF MATTER

A substance can be a solid, a liquid or a gas. States of matter alter with changes in temperature. Whether a substance exists as a solid, a liquid or a gas depends on the material itself and its temperature.

Penny and Das are exploring a range of materials to learn more about states of matter.

1. First they create a chart to record the results of their tests. Using your understanding of each state of matter, complete their table of properties.

State	Has weight 	Has volume 	Has a fixed shape 	Has a fixed volume 
Solid brick	yes			
Liquid oil	yes	yes		
Gas in a balloon			no	

2. Penny and Das investigate compressing (squashing) each substance. Which state or states of matter can be compressed?

3. They investigate changes in the state of water by changing its temperature. Complete the table, indicating whether heating or cooling is needed to make the change occur.

Change of state	Term used to describe the change	What change of temperature is needed? Heating or cooling?
Solid to liquid		
Liquid to gas		
Gas to liquid		
Liquid to solid		
Solid to gas		

4. In your workbook, draw and label a circular diagram, using arrows, to show the information recorded in the table above.



A huge sandhill of solid sand grains can act like a liquid by flowing downhill. Sometimes this is described as a different state of matter. What do you think? Find out more about 'flowing solids' or 'plasma', which are states of matter that behave differently to the usual solids, liquids and gases.

SATURATED SOLUTION



A class of Year 8 students has been growing crystals using saturated solutions. A saturated solution has as much solute as possible dissolved in it.

Emily and Jan decide to create a detailed set of instructions for making a saturated solution of alum (an aluminium potassium sulphate compound).

1. In the 'comment' space next to each step, record something you think about the instruction when you read the step. It could be a question, how clear the instruction is or something you would change.

How to make a saturated solution

Step	Instructions	Comment
1 	Start with a clean jar that can be covered and shaken. Add water to the jar.	
2 	Add an amount of alum to the water. Shake well and let stand for 30 minutes.	
3 	If there is NO powdered alum visible on the bottom of the jar, then repeat Step 2 until there is.	
4 	Carefully pour off the liquid into another clean jar (without getting any powder into the liquid) and seal this new jar tightly. This is your saturated solution.	

2. The instructions do not indicate how big a jar, how much water or how much alum should be added. Do you think this matters?

3. In Step 3, what does it mean if alum is left in the bottom of the jar?

4. When a saturated solution is cooled, some of the solute crystallises out of the solution. How do you think temperature affects crystal growth?



Write and illustrate your own set of instructions for making a saturated solution or some other chemical process.

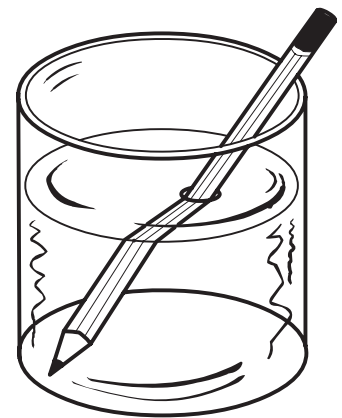


REFRACTING LIGHT

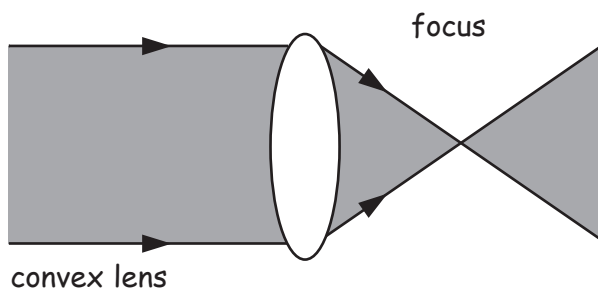
Elinor and Van are investigating how different materials refract light. Refraction is the bending of light as it moves from one material to another, such as from water to air. This happens because light changes speed as it moves from one substance to another. When it enters water from the air it slows, making it change direction.

Elinor and Van observe that a pencil in a glass of water looks broken or bent when they look at it from different angles.

1. Using your knowledge of refraction, explain what makes the pencil look broken.



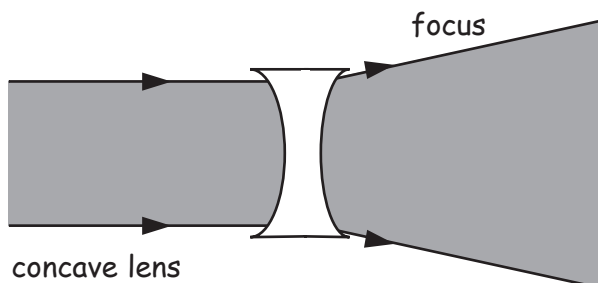
2. A convex lens bends light as it enters and exits the lens.



a. What shape is a convex lens?

b. In what direction does the light change when it passes through a convex lens?

3. A concave lens bends light as it enters and exits the lens.



a. What shape is a concave lens?

b. In what direction does the light change when it passes through a concave lens?



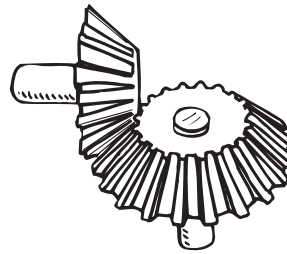
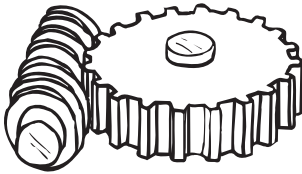
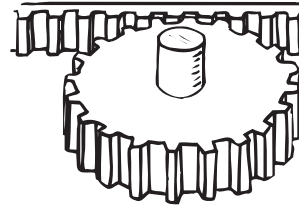
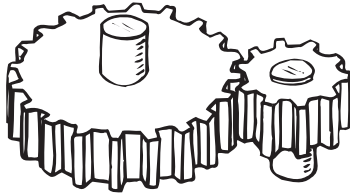
Investigate how refraction is used in making lenses for glasses and magnifiers.

USES OF GEARS

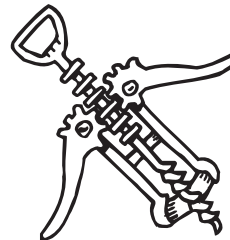
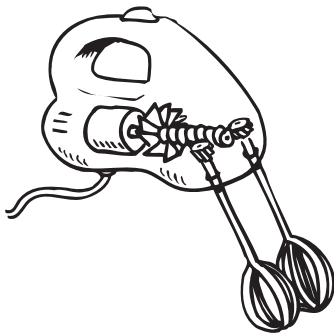
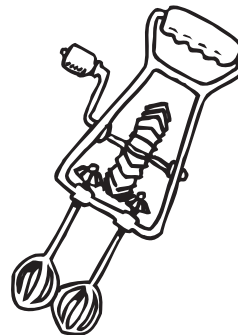
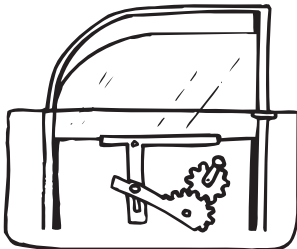


Four basic types of gears are used to transfer movement. Glen and Darren are investigating how each one changes movement in different ways.

1. Help Glen and Darren identify the direction of movement by drawing arrows on the diagrams.



2. Glen and Darren investigate the gears in a number of household items. Label the type of gear in each item and describe the change of movement they create in the device.



3. In your workbook, create a table to show how each of the four types of gear changes movement.



Find some old machines and take them apart to identify their gears.



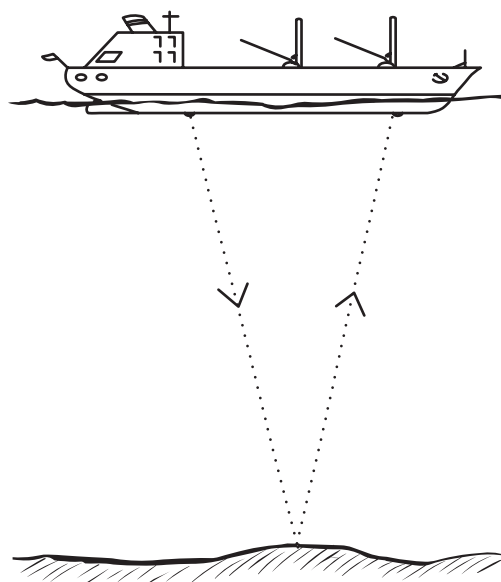
PLOTTING OCEAN DEPTHS

The depth of the ocean can be plotted using sonar. A ship on the surface emits a sound that bounces off the bottom and is picked up by a receiver in the boat. The speed of sound in water is 1450 metres per second. The depth of water can be calculated with the formula: $\text{depth} = \text{speed of sound} \times \text{time} \div 2$.

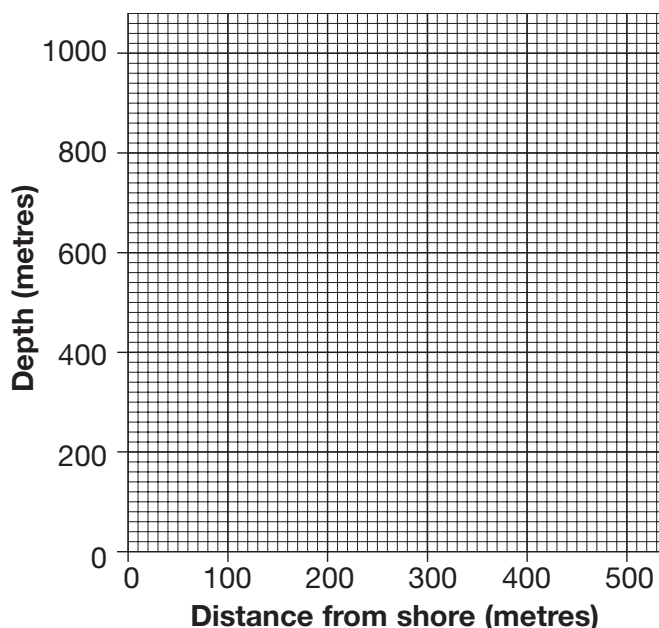
A ship sails from the shore taking readings every 50 metres.

1. Work out the depth of water at each distance from shore, using the formula given in the introduction.

Distance from shore (m)	Time for echo to return (sec)	Depth (m)
0	0.0	
50	0.3	
100	0.4	
150	0.5	
200	0.8	
250	3.5	
300	7.0	
350	8.0	
400	3.6	
450	2.1	



2. Plot the depths on the graph.
3. What does the line on the graph represent?



Sonar stands for sound navigation and ranging. Investigate how people and animals use sonar.

HARDNESS OF ROCKS



In 1812 Frederick Mohs developed the Mohs Scale. This scale is used to define the hardness of minerals. Hardness is determined by a rock's resistance to scratching. Mohs Scale ranks minerals from softest (1) to hardest (10).

Selina and Piri are investigating the hardness of three different rock samples. They decide to compare the hardness of the rocks by hitting the rocks with a hammer.

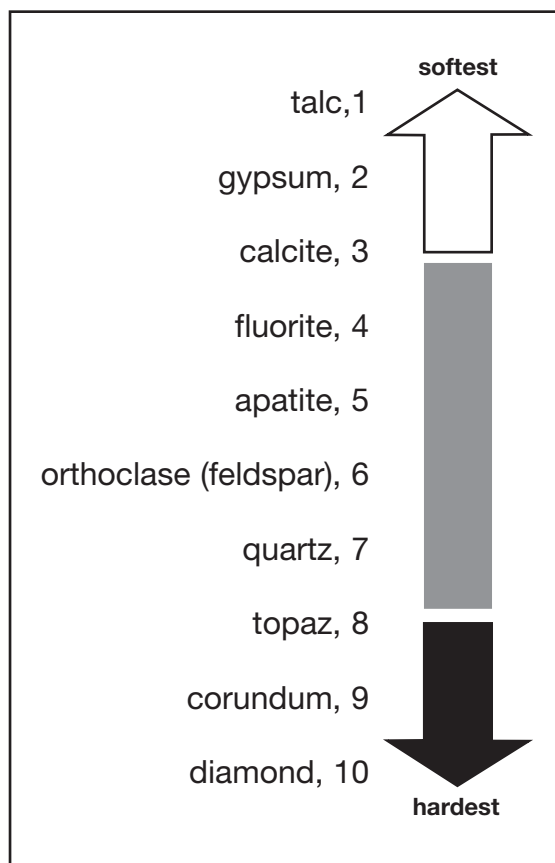
They wonder how to conduct the hardness test safely.

1. Describe how you think their hardness test can be conducted safely.

2. What can they do to make their hardness test fair?

3. What other features of the rocks can they observe or test?

Mohs Scale



4. Help Selina and Piri by creating a chart for them to record their observations.

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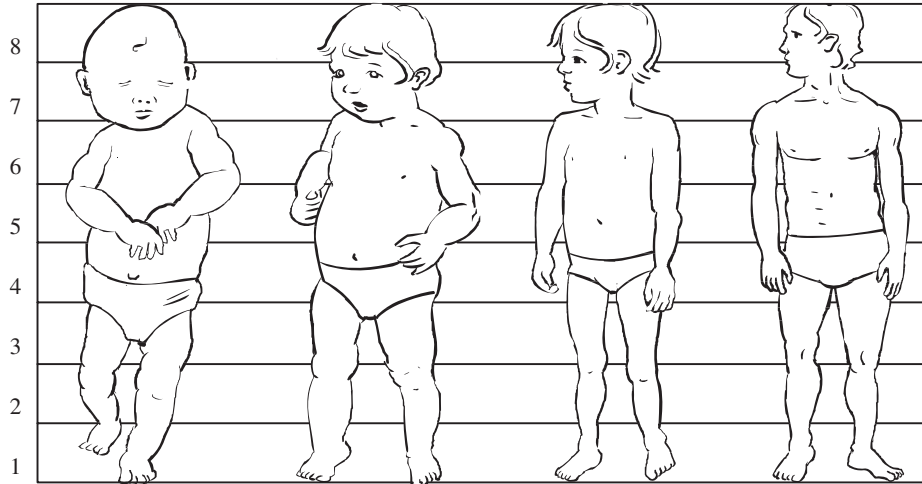
Rock hardness is often tested by seeing if one rock scratches another. How do you think this shows the rock's hardness?



BIG HEAD

Erin observes that her baby brother has a really big head compared with the size of his body. She decides to study the change in body proportions of humans as they grow from baby to adult.

Erin collects measurements from her family and records them on a grid divided into eight equal parts.



1. a. What proportion of the total body is the head of the newborn baby?

b. What proportion of the total body is the head of the adult?

2. a. What proportion of the total height are the legs of the newborn baby?

b. What proportion of the total height are the legs of the adult?

3. Erin's grid does not show actual heights of the people.

a. What are some advantages and disadvantages of using this method to show the information?

b. Do you think it is a good idea?

c. Are there any ways in which Erin's grid is confusing?



Use the information in Erin's grid to graph the change in head size compared with the total body.

BEAKS & FEET

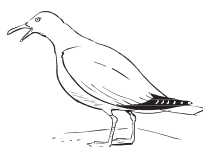


Looking at birds' bodies can provide information about how they live. The feet and beaks in particular are adapted to gaining food in different ways.

Look at the beaks and feet of the five birds illustrated below.



Eagle



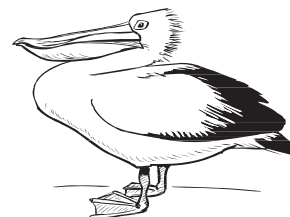
Gull



Parrot



Honeyeater



Pelican

- Fill in the chart of information about their beaks and feet. Use the information in the chart to predict what each of these birds might eat.

Bird	Beak	Feet	Food
Eagle			
Gull			
Honeyeater			
Parrot			
Pelican			

- All birds have feathers and wings but not all birds can fly. In your workbook, use pictures and labelled diagrams to describe how the body of a flightless bird is adapted to its habitat.
- Birds also come in a great variety of colours. What may have led to the evolution of colourful plumage in birds?

- Use the information above to help you explain how structure and function in living things are related.



There are more than 9000 different species of birds with an amazing array of shapes, sizes and habitat. The class Aves (birds) is grouped into 23 different orders. Find out the order name and characteristics of one of the birds investigated above (or another favourite bird species). How are birds grouped into orders?



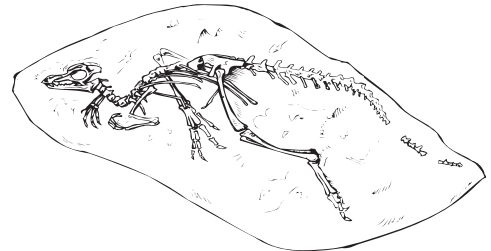
DINOSAURS ALIVE

If dinosaurs were alive today, what would they be like? Dinosaurs were first thought to be extinct reptiles and therefore cold-blooded. (The body temperature of reptiles changes with their environment.) Further evidence meant that ideas about dinosaurs changed dramatically. In 1967 Robert Bakker presented his theory that dinosaurs would have been warm-blooded creatures.

Kim and Louise are presenting arguments to their class that living dinosaurs would have been warm-blooded. Describe how they can use each piece of evidence listed below to support the theory.

1. *Like birds and mammals, dinosaurs had rapid rates of growth.*

How does this statement support the warm-blooded theory? _____



2. *Dinosaurs did not sprawl like most living reptiles.*

How does this statement support the warm-blooded theory? _____

3. *Dinosaur footprints and long limbs show they were capable of high speeds.*

How does this statement support the warm-blooded theory? _____

4. *The range of structures on dinosaurs, such as the sails on the back of the stegosaur, have been identified as controlling body temperature.*

How does this statement support the warm-blooded theory? _____

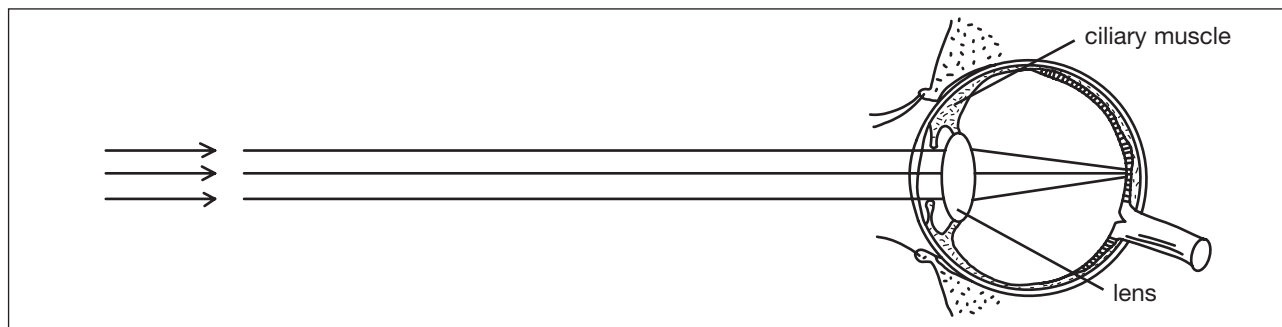


.....
List some different ways that today's living mammals, such as humans, bats, cats, dogs, elephants and whales, control their body temperatures.
.....

HOW DOES AN EYE FOCUS?



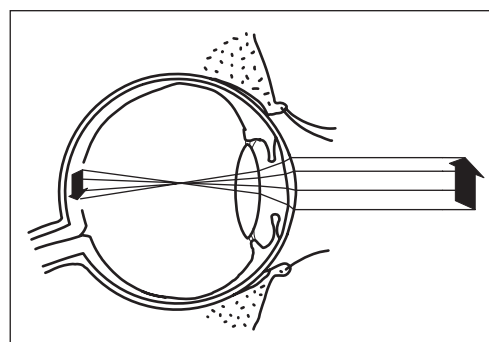
When light passes through the pupil it is focused by the lens onto the retina. The lens shape can be changed by the small ciliary muscles.



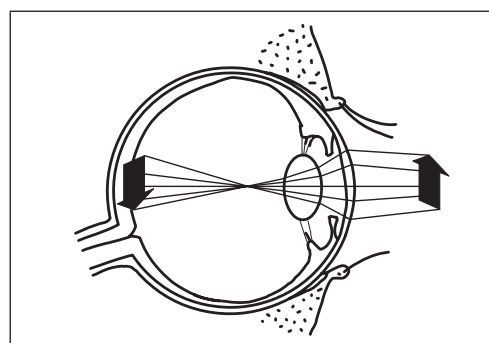
Jenny and Darci are investigating how eyes work to see near objects and far objects. They dissect an eye to identify the parts and then show in a diagram what happens to light rays as they enter the eye.

1. Looking at their drawing above, which part or parts of the eye bend light rays?

2. They create a diagram of how a normal eye focuses on near objects by changing the shape of the lens. How do the muscles make the lens change this way?



3. They create a diagram of how they think a normal eye focuses on far objects by changing the shape of the lens. How do the muscles make the lens change this way?



4. How does changing the shape of the lens affect the path of light?



The term for the ability of a lens to change shape to focus on objects at different distances is 'accommodation'. Design a test to investigate how the ability and speed of accommodation changes as people age.



SURFACE TENSION

A drop of water falling from a tap becomes round due to surface tension. The tension is caused by forces between the molecules in the liquid pulling on each other. The water molecules pull on each other until the droplet has the lowest possible surface area.

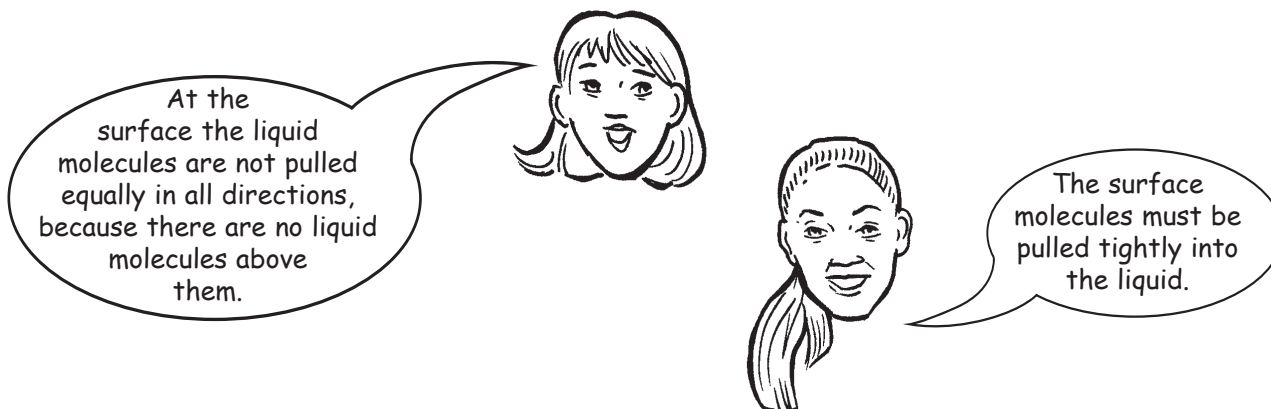
Elsie and Kirra are observing examples of surface tension.

1.
 - a. They carefully add water to a glass until it is filled to the top. When adding more drops, they see the surface bow upwards. Draw what you think they see in the glass.
 - b. They interpret what they see as an example of surface tension. They describe the surface of a liquid behaving as if it is covered in a stretched elastic skin.

Your diagram

2. Next they carefully place a paper clip onto the surface of the water. Describe what you think would happen.

3. They try to explain what happens by thinking about the molecules in the liquid.



Explain what you understand surface tension to be.



Devise a test to find how strong surface tension is.

GEAR RATIOS

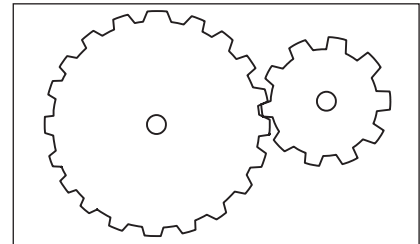


Gears are cogged wheels that turn together. If the wheels are different sizes they will turn at different rates. The one being turned is called the drive wheel. The gear ratio is the number of turns of the drive wheel compared with the number of turns of the wheel being driven.

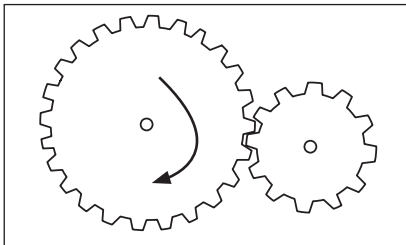
Susan and Graham work out that a gear ratio depends on the number of cogs on the drive wheel compared with the number of cogs on the wheel being driven.

Gear ratio = number of cogs on driving wheel: number of cogs on driven wheel

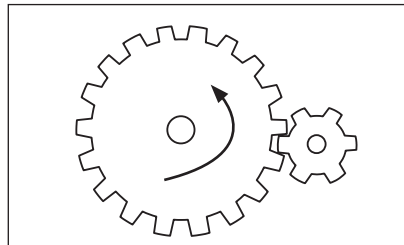
1. On the diagrams below, show which way the driven wheel will turn if the drive wheel is turned in the direction shown. Fill in the gear ratios underneath.



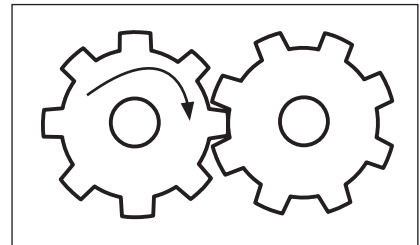
Gear ratio is 2:1



Gear ratio is _____



Gear ratio is _____

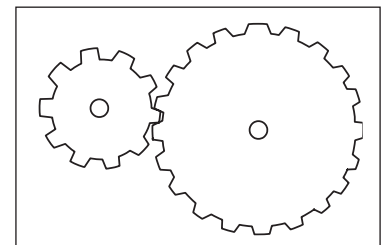
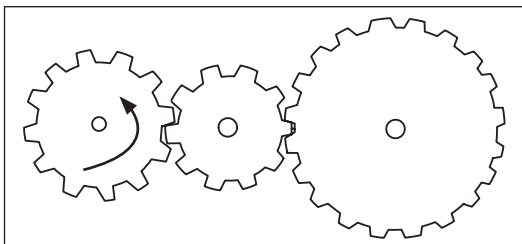


Gear ratio is _____

2. Use the table to record four possible pairs of cogged wheels with a gear ratio of 1:5.

Example 1	Example 2	Example 3	Example 4

3. Which wheel turns faster in a gear with a 1:2 ratio (right)?



4. Predict which direction the wheels will turn when three wheels are together in a system? Draw a diagram in your workbook to help check your prediction.



Design a system with six gear wheels in which the first and the sixth wheel turn at the same rate.

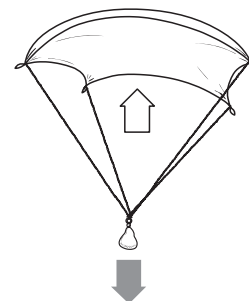


TESTING PARACHUTES



Van and Alicia's class is conducting rate-of-fall tests on parachutes they have made. They were hoping to use their tests to help them to improve their designs. It is easy to see when the weight pulling down is too great for the drag of the parachute. It is harder to reason about the relationship between the features and performance of a parachute that does work.

Van and Alicia think that close observation of parachutes as they fall will help them make design changes. They observe and record details of four factors.



1. **Rate of fall** – *How fast does the parachute fall?*
What does the rate of fall tell you about parachute design?

2. **Flight path** – *Does the parachute fall straight down, at an angle, or swing while descending?*
How does the flight path tell you about what to change in a parachute design?



3. **Inflated canopy** – *Does the canopy remain fully inflated, or does it sometimes partially or fully collapse?*
How does the canopy inflation tell you about what to change in a parachute design?

4. **Stability** – *Does the canopy remain balanced during descent, or does it list to one side, flip over or oscillate while descending?*
How might stability help in assessing parachute design?



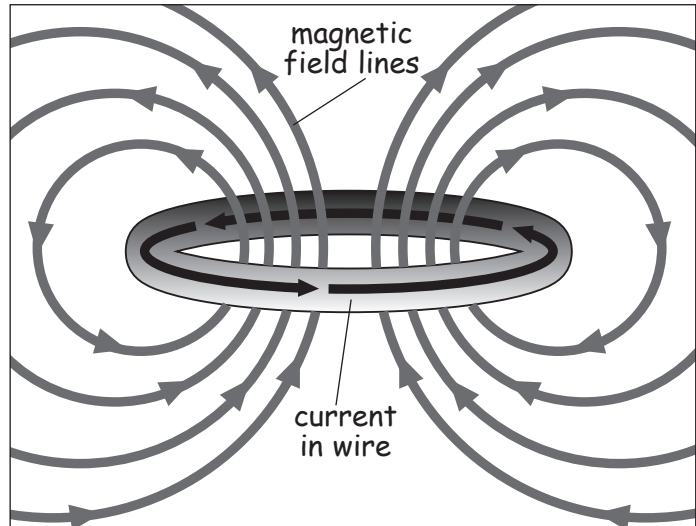
The students are trying to reason about the 'cause and effect' relationships between the parachutes' features and performance. What do you think 'cause and effect' means? How do science experiments help link cause and effect?

MAGNETISM & ELECTRICITY



Hans Christian Oersted predicted in 1813 that there was a connection between electricity and magnetism. In 1819 he placed a compass near a wire carrying an electric current. He saw the compass needle deflected when the electricity was flowing. This was the first time that electric currents were shown to produce magnetic fields.

The magnetic field lines circle around a wire through which an electric current is flowing.

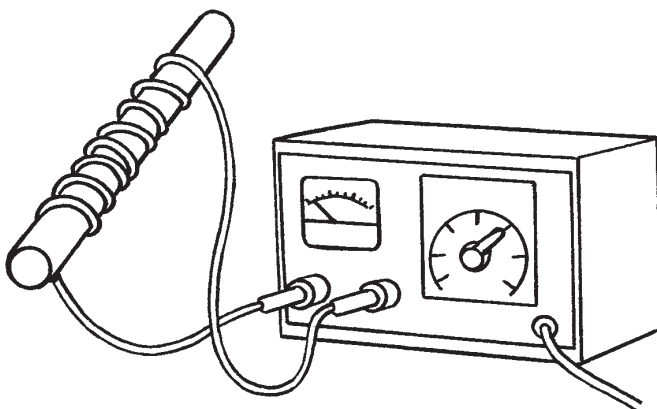


1. What do you understand by the term 'magnetic field'?

2. Describe how you think a compass shows magnetism.

3. Draw small compasses onto the illustration above of the magnetic field around the wire. Show the direction the compass needles might point.

4. Michael Faraday showed that a moving magnetic field could make electricity flow. Using the equipment shown, design a test to show this is true.



How is the link between electricity and magnetism used in making electromagnets or generating electricity?

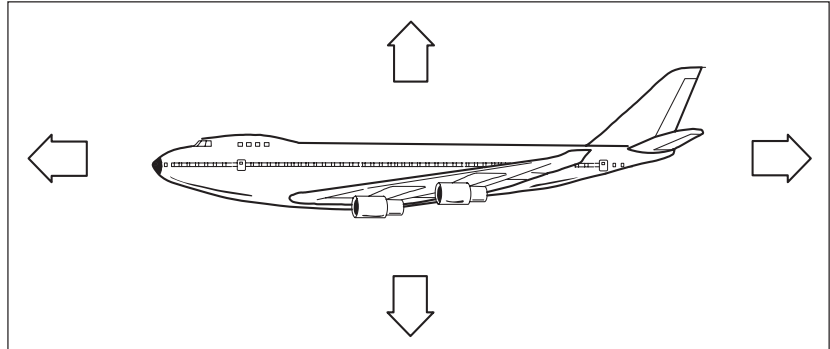


FLIGHT & FORCE

Forces are usually shown using arrows. A plane in flight can be shown as having four forces acting on it: thrust, drag, weight and lift.

Max and Tom are investigating how planes fly.

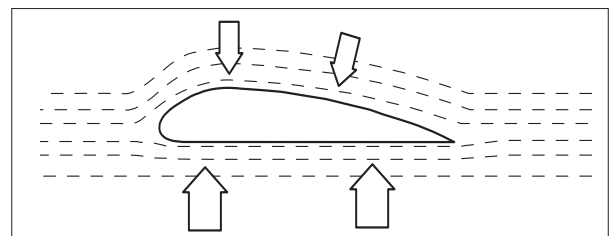
1. Label the four forces on the diagram of a plane in flight.
2. What creates each of these forces?



3. To fly, an object has to overcome the downward pulling force of gravity caused by its weight. How is the upward force that allows a plane to get off the ground created?

4. How does the streamlined shape of a plane help it to fly?

5. Wings are shaped to help planes fly. Use the diagram to help explain how wings function.



6. The different pressure above and below a wing is described as the Bernoulli effect. Daniel Bernoulli was the first to observe it in 1738. You can observe this effect by holding a piece of paper in both hands in front of your mouth and blowing. What happens to the paper and why?



Use your knowledge of forces on a plane to explain why engines have to work harder for a plane to take off than when they travel along at one height.

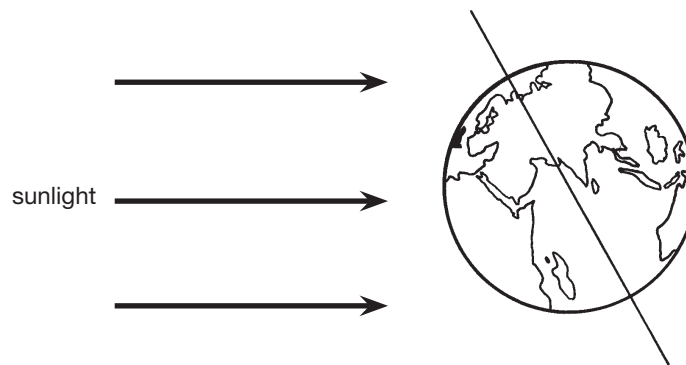
SEASONS & SUNLIGHT



Different parts of the Earth have different seasons during the year because the Earth is tilted on its axis. This tilt causes changes in the angle of the Sun in the sky and the number of hours of sunlight at different times of the year.

Erica and Gemma are trying to explain how the temperature changes at different places on Earth during the year.

1. They try to imagine standing at one of the poles. Write labels onto this diagram to help explain how the number of hours of daylight changes through the year.



2. How do you think the number of hours of sunlight would affect the temperature?

3. How does the angle of the Sun vary at the Equator compared with close to the poles?

4. Use the diagram to help explain how the angle of the Sun affects temperatures.



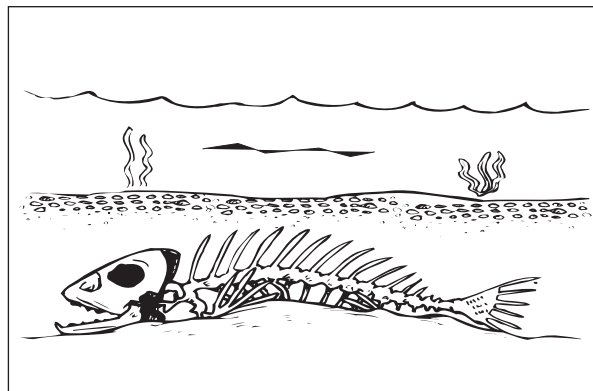
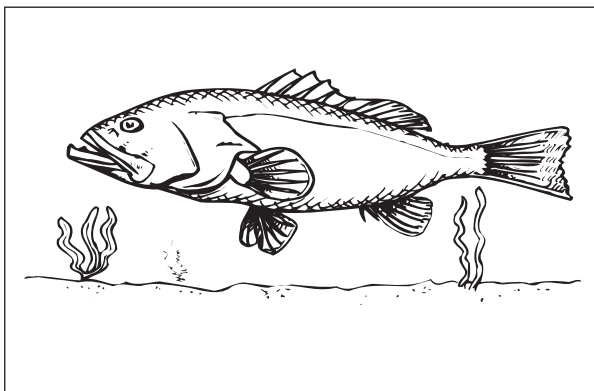
Some of Erica and Gemma's classmates think that the seasons are caused by a change in distance between the Earth and the Sun at different times of year. Use a model of the solar system to show this is incorrect.



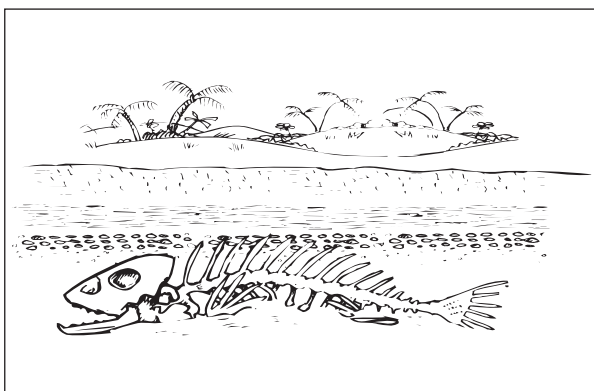
HOW FOSSILS FORM

Rocks sometimes contain the fossilised remains of things that lived long ago. Geologists conclude this is evidence of continuous change of the Earth's surface. The most common fossils are remains of marine organisms.

Helen and Toby use a series of diagrams to show how fossils formed and became exposed. They describe what happens at the first stage.



Dying fish falls to the bottom of the sea.



1. Using the space under the drawings, help Helen and Toby describe the next three stages of fossil formation.
2. How does this series of drawings help explain the fact that fossils are most often found in sedimentary rock?



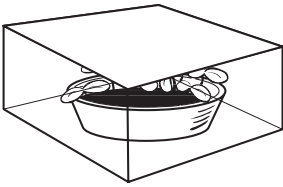
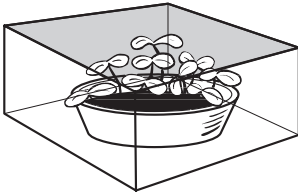
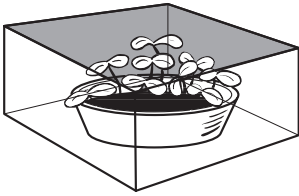
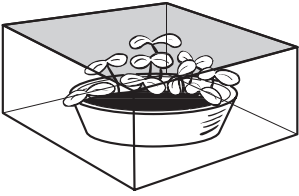
What parts of animals are most likely to be preserved as fossils?

LIGHT & GROWTH



Danny and Lil are investigating how the colour of light influences plant growth.

They set up four identical containers, each with eight pea plants. They put each one into a different light condition.

Container 1	Container 2	Container 3	Container 4
closed box (no light)	a box covered with red cellophane	a box covered with blue cellophane	a box covered with green cellophane
			

The diagrams show how Danny and Lil set up their experiment.

1. What are they trying to find out?

2. Suggest two things Danny and Lil need to control to make it a fair test?

3. What conclusions can they draw from their test?

4. Their teacher suggests they set up another two boxes. What light conditions might they use in these?



What was the purpose of having eight plants in each container rather than one?

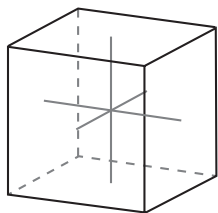


THE SHAPE OF SALT

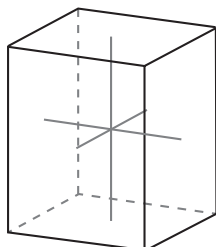
A class of Year 8 students has grown crystals of common salt (sodium chloride). They are trying to work out the shape of the crystals. They know a crystal is a solid with a regular internal structure.

Their teacher gives them the names and arrangements of the five main crystal shapes.

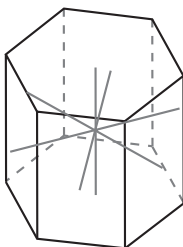
Isometric or cubic



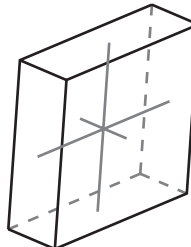
Tetragonal



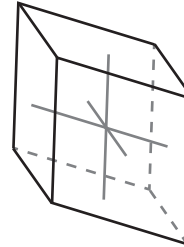
Hexagonal



Monoclinic



Triclinic



1. What is the difference between an isometric crystal and a tetragonal crystal?

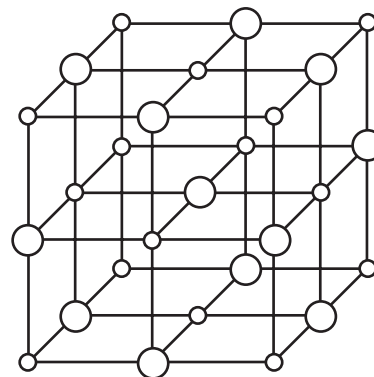
2. In what ways are cubic, tetragonal and hexagonal crystals alike?

3. How are monoclinic crystals different from cubic, tetragonal and hexagonal crystals?

4. How are monoclinic and triclinic crystals similar?

5. The students examine the crystals they have grown. Although the crystals come in different shapes and sizes, they have the same 'arrangement'. Explain how this occurs.

6. The drawing shows how the atoms in common salt (sodium chloride) crystals are arranged. Knowing that crystals form smooth external surfaces (or faces) because of the arrangement of atoms, what shape will sodium chloride crystals take? _____



Examine some sodium chloride crystals under a microscope to check the shape they actually have.

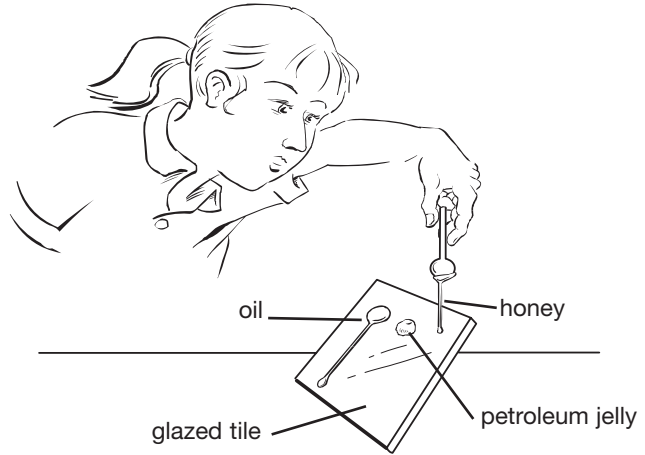
VISCOSITY OF LUBRICANTS



Lubricants are substances such as oil that are applied to machinery surfaces to reduce friction between moving parts. Fluids that are useful as lubricants do not change in viscosity very much when they are heated.

A class uses the equipment shown to conduct tests on viscosity of a number of liquids.

1. How does this equipment test viscosity and what needs to be recorded?



2. What needs to be controlled to make this a fair test of viscosity?

To test the usefulness of the liquids as lubricants, the class decides to heat and cool the tile surface before carrying out the tests.

3. How does heating and cooling the tile make this a test of fluid as a lubricant?

4. What can they do to make this a fair test?

5. How else can they test the change of viscosity with temperature?



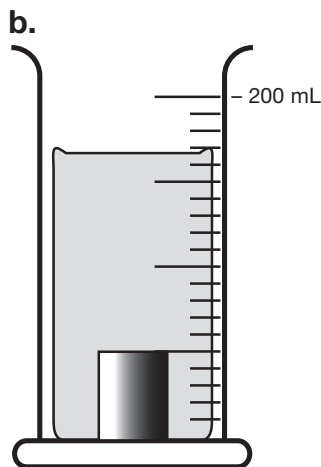
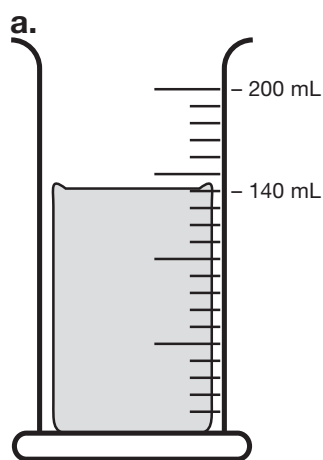
What would you see if you dropped a marble into a jar of honey and a jar containing oil? How could this be used to measure viscosity of fluids?



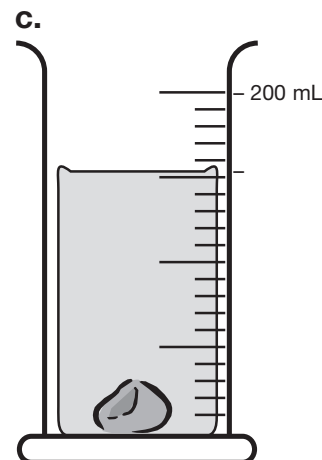
MEASURING VOLUME

The volume of liquids is measured using a measuring cylinder. The volume of regular-shaped solid objects is found by multiplying the length by the width by the height. Measuring the volume of irregular shaped solids requires other methods.

Ian and Ken develop a method to measure the volumes of a range of irregular shaped objects by placing them in water in a measuring cylinder. See diagram **a** for the amount of water they use.



Volume _____



Volume _____

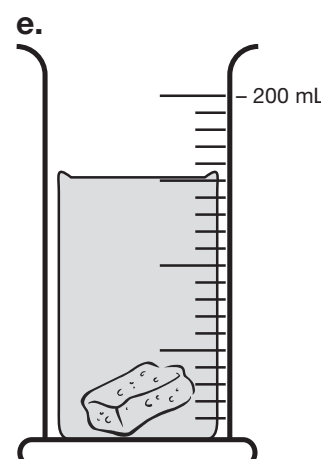
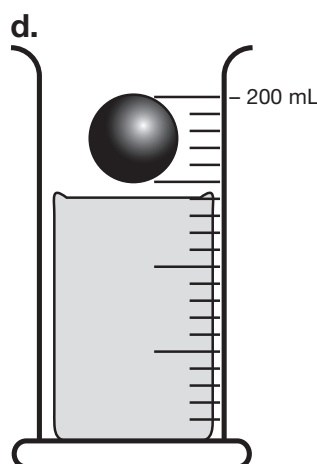
1. Record the volume measurements for diagrams **b** and **c**.

2. A ball is put into the water. What can you do to find out its volume?

3. A piece of limestone riddled with airholes is placed in the cylinder to measure its volume. The water level rises to 155 millilitres but later drops to 151 millilitres. What makes the level drop?

Record the limestone's volume under diagram **e**.

4. What are some problems with the displacement method of finding volume?



Volume _____



What can the displacement method tell you about the density of materials?

ELECTROMAGNET STRENGTH



Joe and Huang are investigating the strength of electromagnets made in different ways. They make 10 different electromagnets; all with the same iron core. They test the magnets' strength by counting how many paper clips they pick up.

They create this table with their data.

Electromagnet number	Number of turns in coil	Voltage (volts)	Number of paper clips picked up
1	10	1	1
2	10	2	
3	10	3	
4	20	1	2
5	20	2	4
6	20	3	10
7	20	4	19
8	20	5	19
9	20	6	
10	30	2	6

1 In your workbook, draw a graph of the results for electromagnets with 20 turns in the coil. How does the voltage affect the magnet strength?

2 How many paperclips will electromagnets 2 and 3 pick up? Write your predictions in the table.

3 What are three things that can affect the strength of an electromagnet?

4 Is it true to say that the higher the number of turns, the stronger the electromagnet?

5 Joe and Huang wonder if the results for electromagnets 7 and 8 are correct. If they are, how would you explain the result?

6 Predict the result for electromagnet 9. Explain your reasoning.

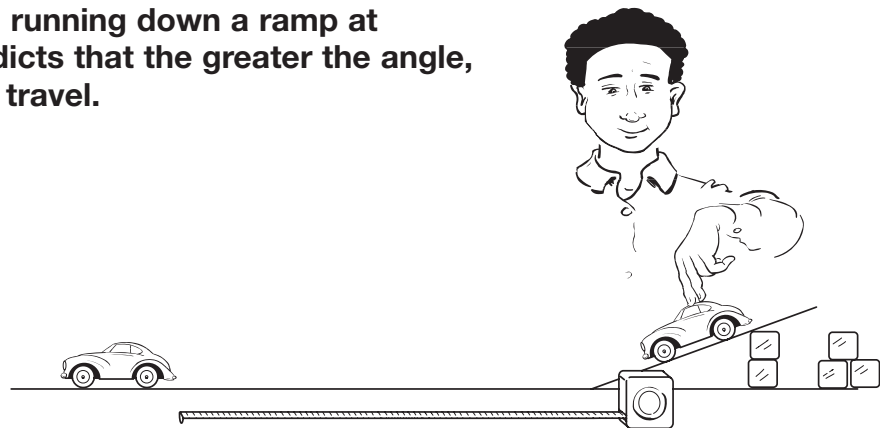


Joe and Huang also find that when the electricity is switched off, the paper clips drop from the iron core. But if they use a steel bar, the paper clips do not immediately fall. Use this information to explain why iron is used for industrial electromagnets rather than steel.

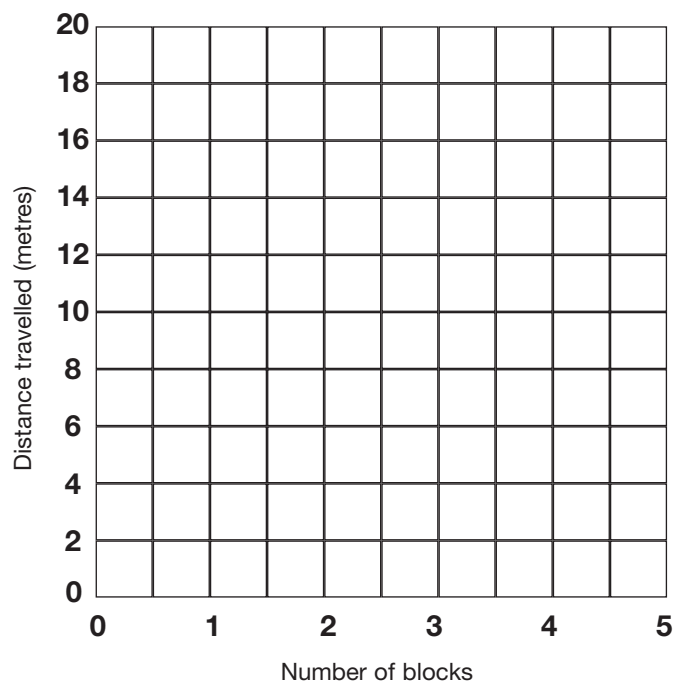


TOY CAR RAMP

Gyula is testing toy cars running down a ramp at different angles. He predicts that the greater the angle, the further each car will travel.



Height of ramp (number of blocks)	Distance travelled (metres)
1	0
2	3
3	4.3
4	16
5	18



1. Plot a graph of his results
2. Decide whether the pattern is a straight line or a curve.
3. How far will the toy car travel if the ramp is at six blocks' height?

Gyula then tests another car that is twice the size of the first. He wonders how the different sized cars will compare.

4. Will the car travel a shorter or longer distance before it stops?
5. What are some other factors that may influence how far the car travels? Explain how they may influence it.

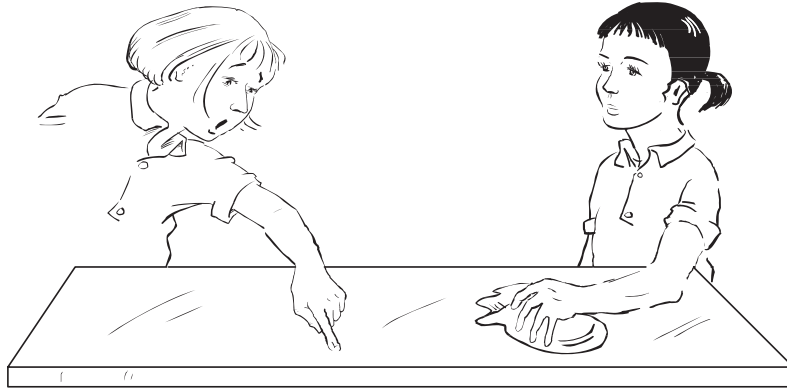


Friction between the car wheel and the surface will slow the cars down. How could you change the surface to make each car go further?

MICROBE IMAGE PROBLEM



A Year 8 class is studying microbes. They discover that only a small number of the wide range of microbes in the environment are dangerous to people. They then watch a television commercial for a kitchen 'anti-bacterial' spray.



1. What do you think the word 'microbe' means?

2. List some organisms that are examples of microbes.

3. What are some problems with people using 'anti-bacterial' substances regularly in the kitchen?

4. Name some helpful microbes.

5. Many people think that all microbes are dangerous to people. List some reasons they may think this.



Create a TV advertising slogan or jingle for 'good' microbes.



WHAT IS A DINOSAUR?

When the first fossil evidence of dinosaurs was uncovered, Sir Richard Owen coined the name dinosaur, which means 'terrible lizard'. He could see these ancient remains had similarities to lizards.



Early reconstruction of an Iguanodon



Modern reconstruction of an Iguanodon

1. In what ways are the two reconstructions similar?

2. In what ways are they different?

3. What might have led to ideas about dinosaur structures changing?

4. Will we ever know exactly what the Iguanodon was like in real life? Why?



In science fiction movies, dinosaurs are often shown as brown and roaring. What evidence gives scientists ideas about what skin and skin colour dinosaurs had and what sounds they made?

WHAT IS ENTOMOLOGY?



People who speak Latin know the word entomology means 'the study of insects'. Many words used in science are built from Latin roots.

Benny and Tim use the Latin beginnings and endings in the table below to work out the meanings of some terms used in biology. For example, they find the meaning of entomology by looking up *entom* + *logy* in the table.

- *entom* means insect
- *logy* means the study of
- *entomology* is the study of insects

Beginning (prefix)	Meaning	Ending (suffix)	Meaning
bio	life	ata	characterised by
centi	hundredth	cules	small
cephalo	head	logy	study of
entom	insect	ptera	wing
lepid	scale	pod	foot
meta	between	sis	condition, state
morph	shape, form	ped	foot

Work out the meaning of these science words.

1. Biology

bio = _____ + logy _____

Biology is _____

2. Metamorphosis

meta = _____ + morph _____ + sis _____

Metamorphosis means _____

3. Lepidoptera

lepid = _____ + ptera _____

Lepidoptera means _____

4. Based on the meaning of the word and your own knowledge of insects, what group of insects is the order known as Diptera? The ending *ptera* means wing but the other part of the word *di* is not in the table.

In your workbook, list some other words that begin with *di*. How does this help you decide on the meaning of 'diptera'?

5. Look up some other insect order names in the dictionary such as coleoptera, hemiptera and arthropod. You can use these meanings to add to the table.



Make up some names of imaginary insect groups using some of the Latin terms.



SOUND & LIGHT IDEAS

The Ancient Greek philosopher Aristotle (AD 384–322) conducted thought experiments to learn more about the world. One thing he thought was that sound and light travelled through air just like waves in the sea. He then concluded that light and sound was not able to travel without a material to carry the waves.

1. What may have led Aristotle to these ideas?

2. In what ways are light and sound like waves in the sea?

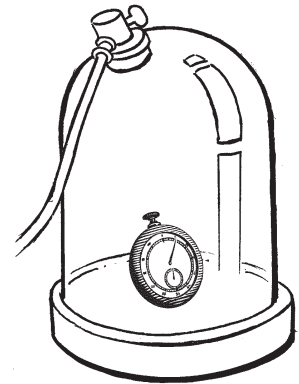
3. In what ways was Aristotle right? What evidence do you have to support your ideas?

In the 1640s Evangelista Toricelli (1608–1647) succeeded in making a vacuum. It was then shown that Aristotle was wrong about light. A candle could be seen through a jar with a vacuum, therefore light could travel through a vacuum. It was not like water in the waves of the sea – it did not need a material to carry it.

4. What do you understand by the term ‘vacuum’?

5. What other evidence supports the idea that light does not need material to travel?

Robert Boyle (1635–1703) showed that sound could not travel through a vacuum. He placed a loudly ticking watch in a bell jar and then pumped out all the air. As the air was taken out, the sound of the watch ticking became fainter and fainter until it could not be heard at all. As he let the air back in, the sound of the ticking returned.



6. Explain why the ticking sound became fainter as the air was pumped out.

7. What did Boyle’s experiment show about how sound travels?

8. In space movies, spaceships are often shown flying past with a loud whooshing sound. What is wrong with that?



.....
Aristotle’s idea was based on thinking, not experiments. Was he a scientist?
.....

READING ELECTRIC CIRCUITS



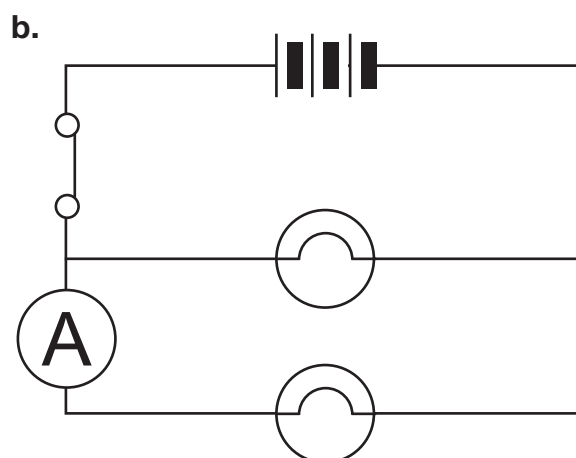
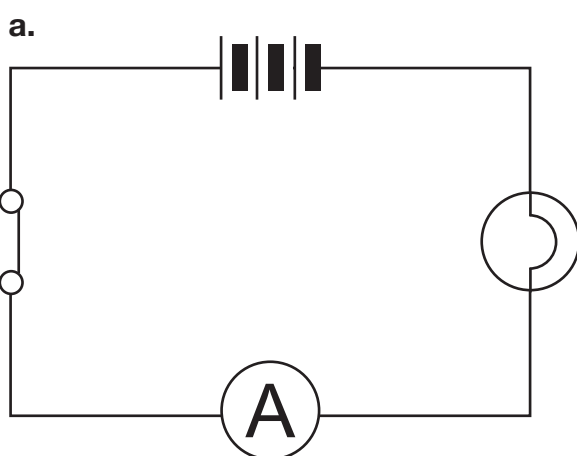
Standard symbols are used to draw diagrams of electrical circuits. To understand diagrams of electrical circuits, Ruth and Nat need to be clear about what the symbols mean.

Use the following symbols to help Ruth and Nat with their circuit diagrams.

1. In the space in the table, write the name of the item that each symbol represents.

2. Draw a circuit that represents an open switch, a resistor and a battery connected in series.

3. Describe in words the electrical circuits shown in these circuit diagrams.





The two circuits shown above are described as being in series and in parallel. What are some differences between series and parallel circuits?

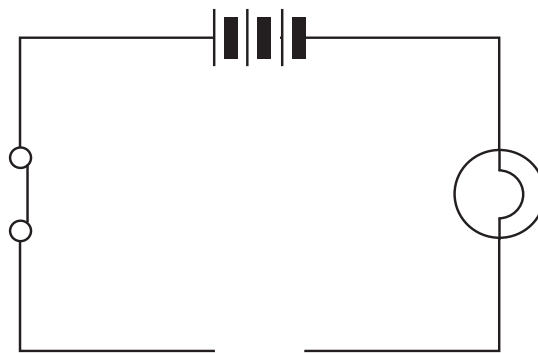


CONDUCTORS & INSULATORS

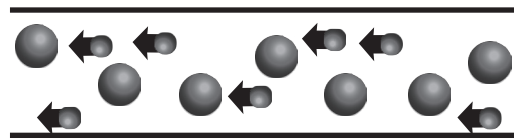
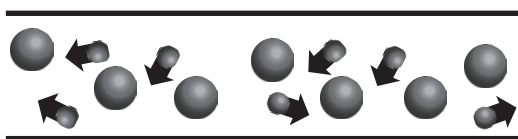
Jane is investigating materials to see if they are conductors or insulators of electricity.

She makes a testing circuit that looks like this.

When she connects a conductor in the gap in the circuit, the lamp lights up. She then tests a range of materials in her circuit tester.



1. Name five things you predict can be put in the circuit and the lamp will light up.
 1. _____
 2. _____
 3. _____
 4. _____
 5. _____
2. Name five things you predict can be put in the circuit and the lamp will not light up.
 1. _____
 2. _____
 3. _____
 4. _____
 5. _____
3. Complete the sentences to show what you understand by the terms 'conductor' and 'insulator' when talking about electricity.
 - a. An electrical conductor is _____
 - b. An electrical insulator is _____
4. In your workbook, use the terms conductor and insulator to explain the predictions you made in questions 1 and 2.
5. Jane drew two diagrams to show what she thought was happening in the wires in her circuit. Label the diagrams to show if they represent the wire when a conductor or insulator was in the circuit.



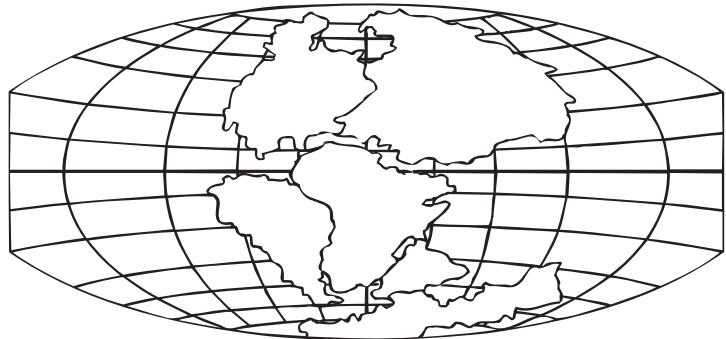
.....
List some items that make use of a material being an electrical insulator.
.....

CONTINENTAL DRIFT IDEAS



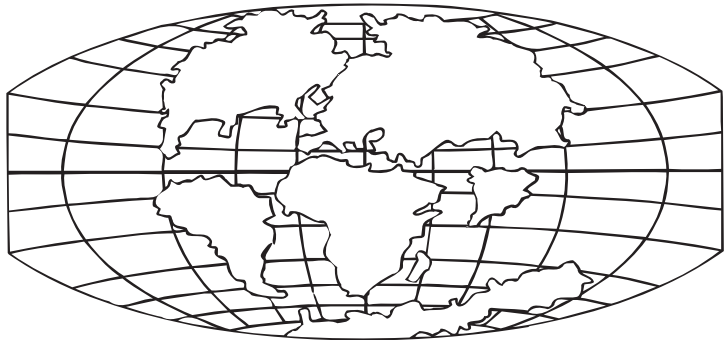
In 1620, the Englishman Francis Bacon drew attention to the remarkable similarity in shape between the west coast of Africa and the east coast of South America.

1. What does the term continental drift mean?



2. How did Bacon's observation support the idea of continental drift?

3. Why do you think Bacon never actually suggested that the two continents had once been joined together?



In 1912 the German meteorologist Alfred Wegener found fossils of tropical ferns on an Arctic island. This led to him developing the Continental Drift theory in 1915.



4. How did finding a fossil lead Wegener to suggest that the continents had not always been shaped as they are today?

5. The idea of continental drift became widely accepted by geologists in the 1960s and 1970s. What evidence could have supported this proposition?

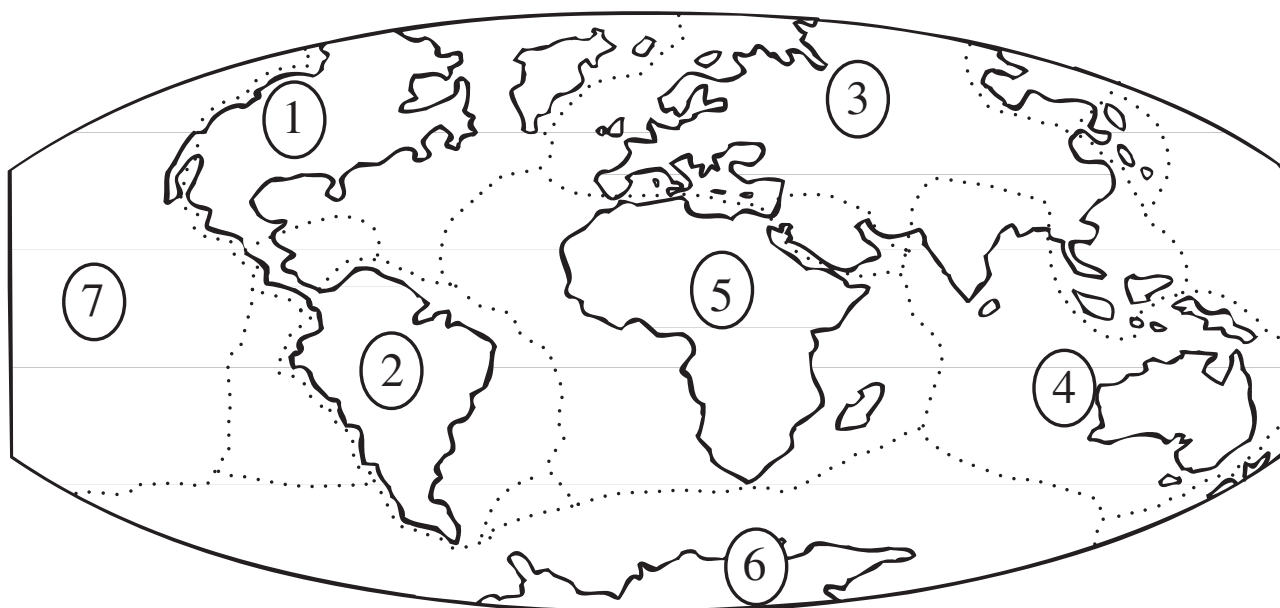


Draw arrows on the diagrams to show the direction that each continent has been moving over the past few hundred million years.



EARTH'S TECTONIC PLATES

Today the Earth's crust is made up of about 7 major tectonic plates and at least 12 smaller ones. The surface of the Earth can be described as a cracked eggshell.



1. In what way is the Earth like a cracked eggshell?

2. What is inside the Earth's cracked shell?

3. What happens when movements cause tectonic plates to collide?

4. How does the Tectonic Plate theory help explain volcanoes, earthquakes and tsunamis?



Use the directions the plates are moving to predict what the Earth's continents might look like 50 million years from now.

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Scientific Enquiry Skills is designed to help students develop skills in the processes of scientific investigation. These activities are especially useful for focusing attention on particular aspects of scientific enquiry. The activities will complement and support any existing science program and are divided into the areas of the living world (biology), matter (chemistry), the physical world (physics), Earth (geology) and space (astronomy). They are further organised into the major investigative processes that students will practise.

The activity sheets include introductions, instructions and questions along with spaces for students to record answers to questions, create graphs or complete data tables. Most activity sheets also include extension ideas. Each activity sheet is self-contained so that students can use them with minimal support from the teacher, working independently, in pairs or in small groups. Many activities centre on discussions between student characters to introduce an issue or problem for students to explore.

Scientific enquiry skills explored

- Choose and use appropriate equipment safely
- Turn ideas into a form that can be investigated
- Make appropriate observations to get reliable results
- Consider key factors in planning investigations and collecting evidence
- Use scientific knowledge to explain and interpret data and make predictions
- Develop shared understanding of scientific terminology
- Identify patterns in data and suggest improvements to tests

Topics include

- Flammability
- Gears
- Electric circuits
- Balancing forces
- Light and sight
- Fluids and temperature
- Magnetism
- Nature of substances
- Reflection of sound

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